

Power, Constant, and Sum Rules

Date_____ Period____

Differentiate each function with respect to x .

1) $y = 6$

2) $y = -8$

3) $y = 7$

4) $y = x^5$

5) $y = 5x^9$

6) $y = -8x^4$

7) $y = 5x^7 + 4x$

8) $y = x^3 - 8x^2$

9) $y = 4x^6 - 7x^5 - 9x^3$

10) $y = 8x^7 + 5x^5 + 9x^3$

Critical thinking question:

11) a , b , and c are constants. If $y = x^{3b} + 3x^{4c}$, what is $\frac{dy}{dx}$?